

FINAL PROJECT REPORT

Software Installation Request Automation

1. INTRODUCTION

1.1 Project Overview

The Software Installation Request Automation project is a comprehensive ServiceNow application designed to digitize and manage the lifecycle of employee software requests. It replaces manual, disjointed email processes with a centralized Service Catalog item.

1.2 Purpose

The primary purpose is to streamline IT operations, ensure compliance with software licensing, accelerate manager approval times, and provide end-users with real-time tracking of their software requests.

2. IDEATION PHASE

2.1 Problem Statement

Employees face significant delays when attempting to acquire necessary licensed software due to a reliance on unstructured emails, lost manager approvals, and manual tracking by IT staff.

2.2 Empathy Map Canvas

The empathy map illustrates the frustration of end-users regarding the lack of transparency in the current IT procurement process.

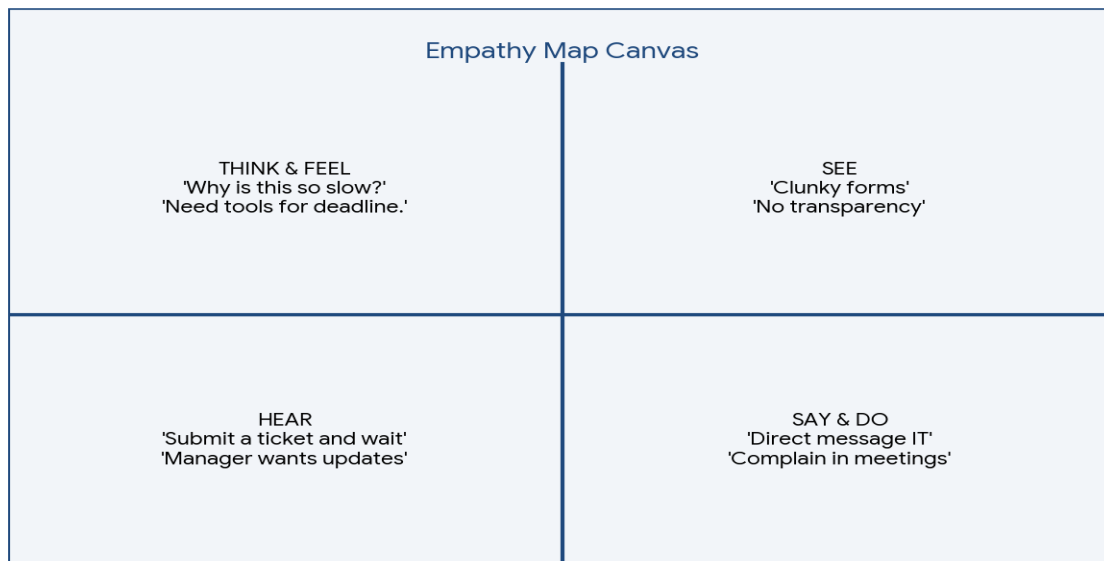


Figure: Empathy Map Canvas

2.3 Brainstorming

Ideas generated included utilizing the ServiceNow Service Portal for intake, leveraging Flow Designer for automated manager approvals, and utilizing Server-Side scripting to monitor and flag license shortages dynamically.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

The user journey simplifies from hunting down email addresses to selecting variables on a structured portal and awaiting automated task fulfillment.

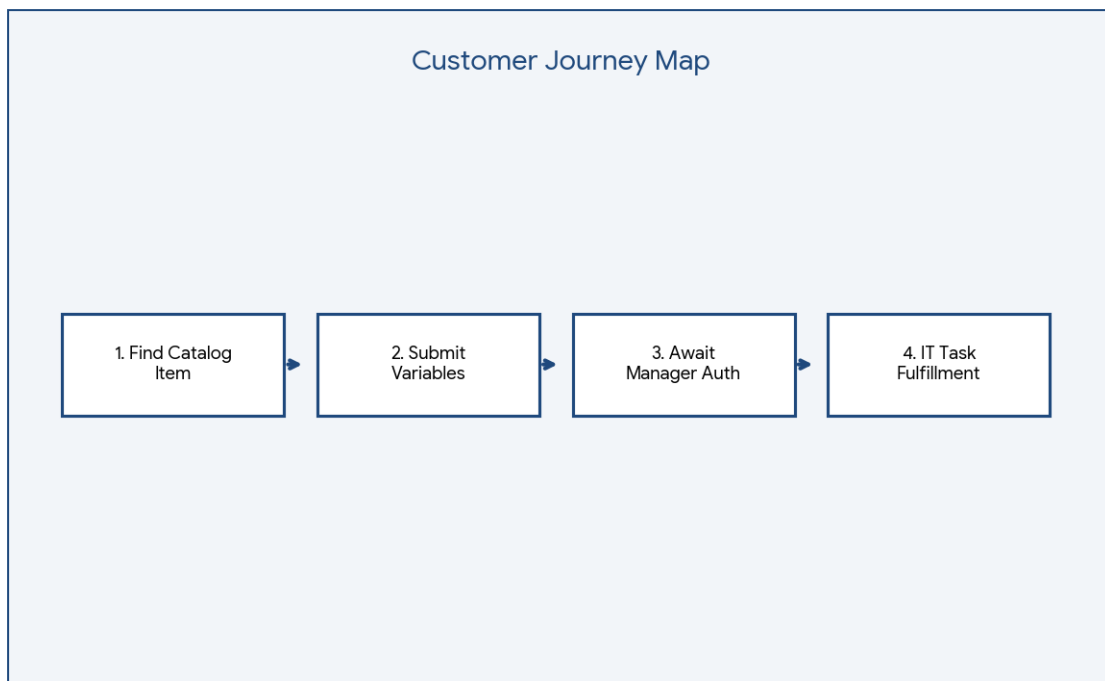


Figure: Customer Journey Map

3.2 Solution Requirement

The solution requires a Catalog Item with four variables (Software Name, Version, Justification, Urgency), an automated Flow to route requests to the 'Software Support' group, and a Business Rule for exception management.

3.3 Data Flow Diagram

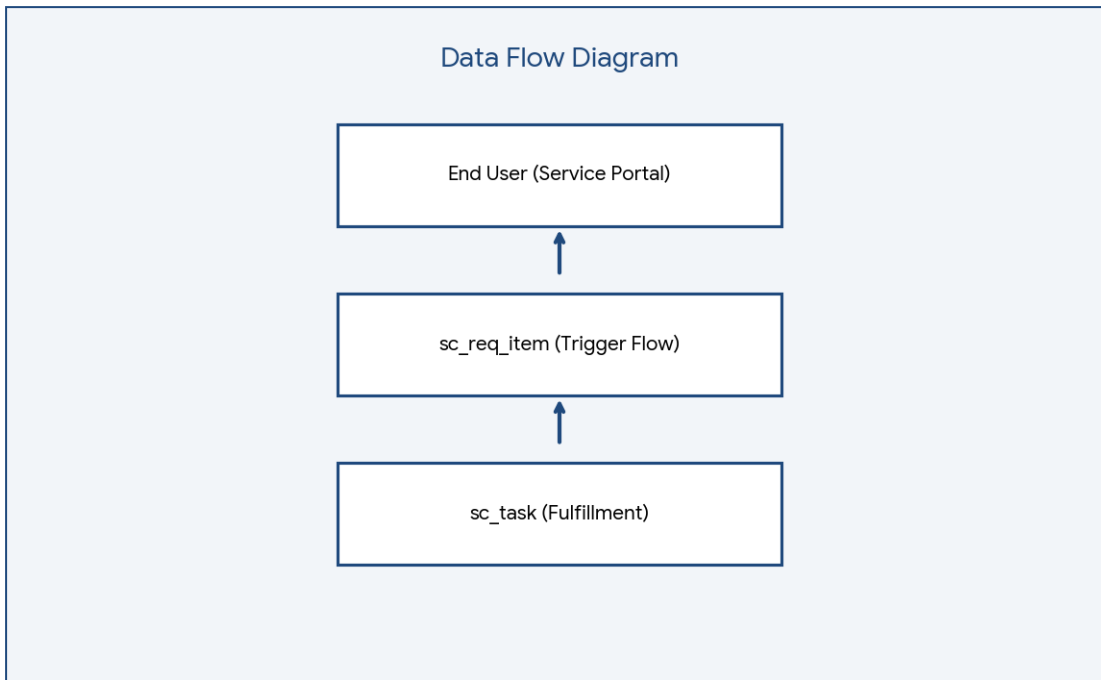


Figure: Data Flow Diagram

3.4 Technology Stack

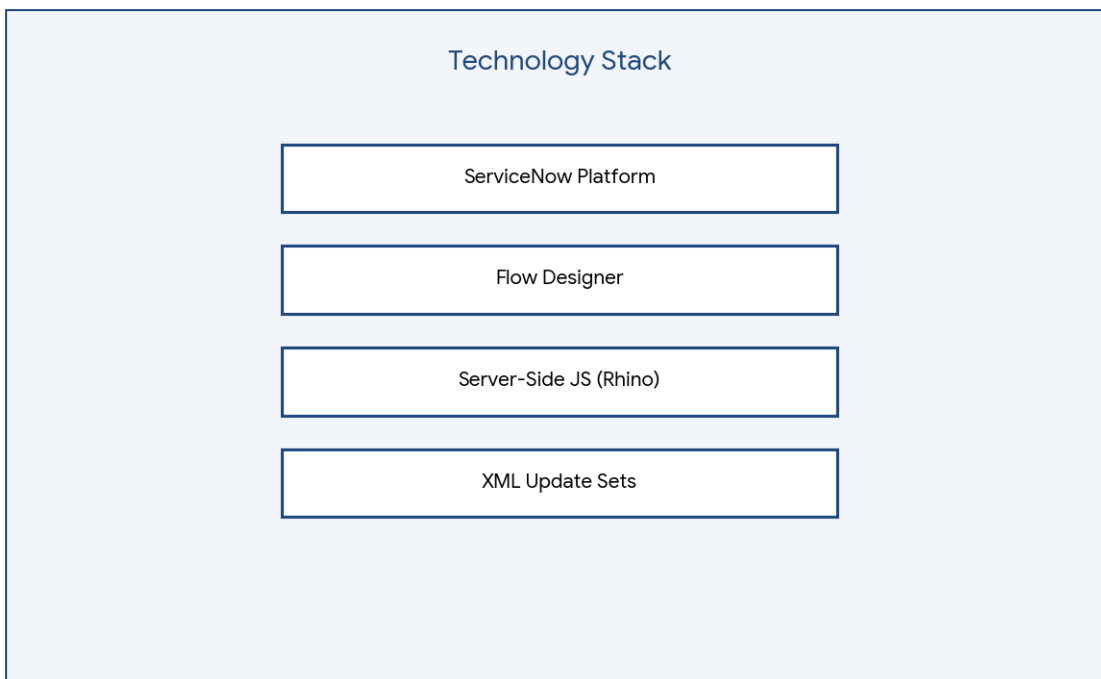


Figure: Technology Stack

4. PROJECT DESIGN

4.1 Problem Solution Fit

The automated Flow Designer fits the problem perfectly by removing the human bottleneck of routing emails and verifying manager authorizations manually.

4.2 Proposed Solution

A scoped ServiceNow application containing a Service Catalog Item. When submitted, a Flow asks the Software Support group for approval. If approved, a Catalog Task (SCTASK) is created. If rejected, the Request Item (RITM) is closed.

4.3 Solution Architecture

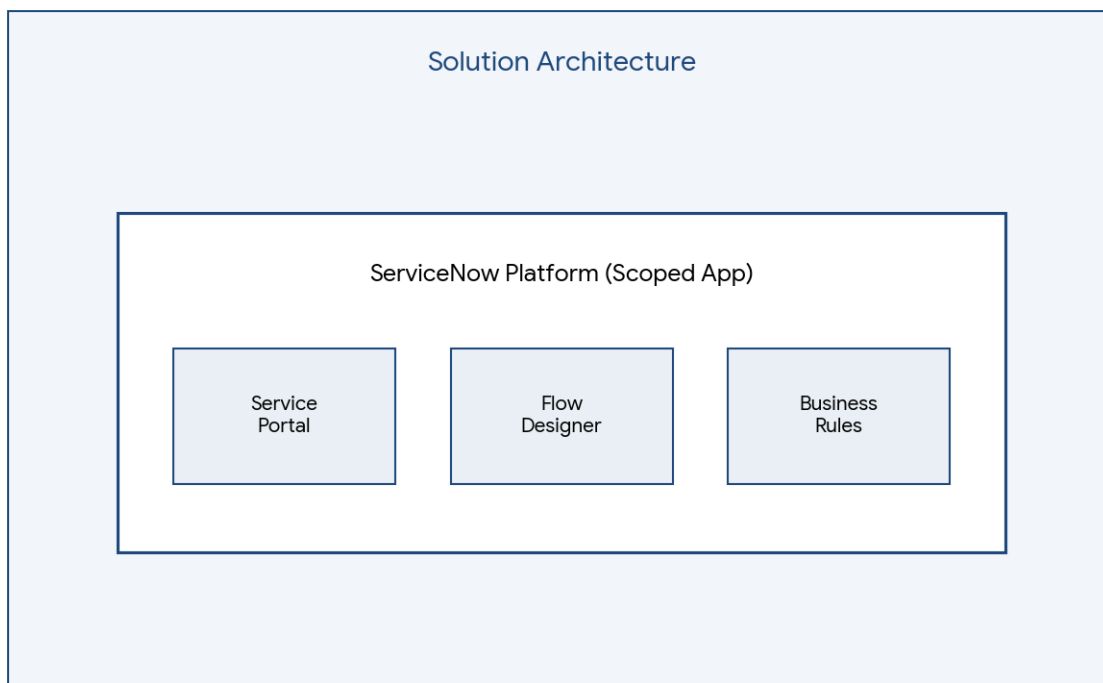


Figure: Solution Architecture

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was executed in an Agile framework consisting of Sprints focusing on UI, Flow Logic, and Scripting respectively.

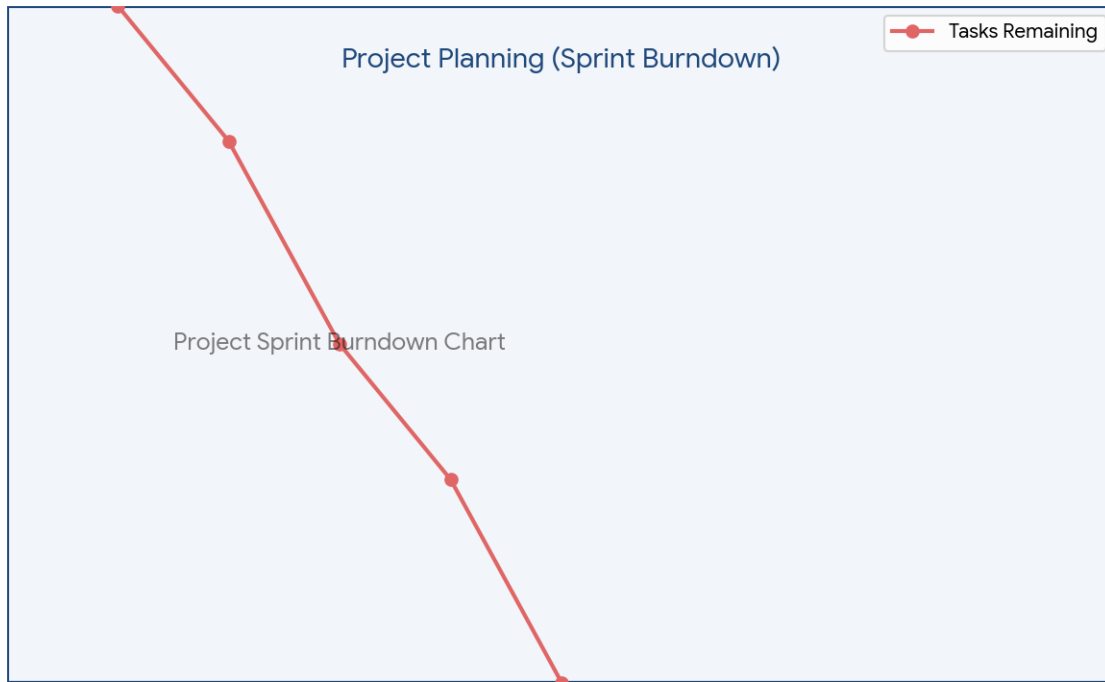


Figure: Project Sprint Burndown Chart

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The application was tested across three primary paths: 1) Happy Path (Successful approval and task creation), 2) Rejection Path (Clean closure of ticket), and 3) Exception Path (Manual state change to 'Pending' successfully generating a priority Incident). All database inserts executed within acceptable platform limits.

7. RESULTS

7.1 Output Screenshots

servicenow

KnowledgeCatalogRequestsSystem StatusCartToursSystem Administrator

Home > All Catalogs > Service Catalog > Software > Software Installation Request

Search Catalog

Software Installation Request

Software Installation Request

What software do you need?

Specify version (if required)

Why do you need this software?

Select urgency level

Critical

Add attachments

Choose a file or drag it here.
Copy and paste clipboard files here.

Quantity: 1

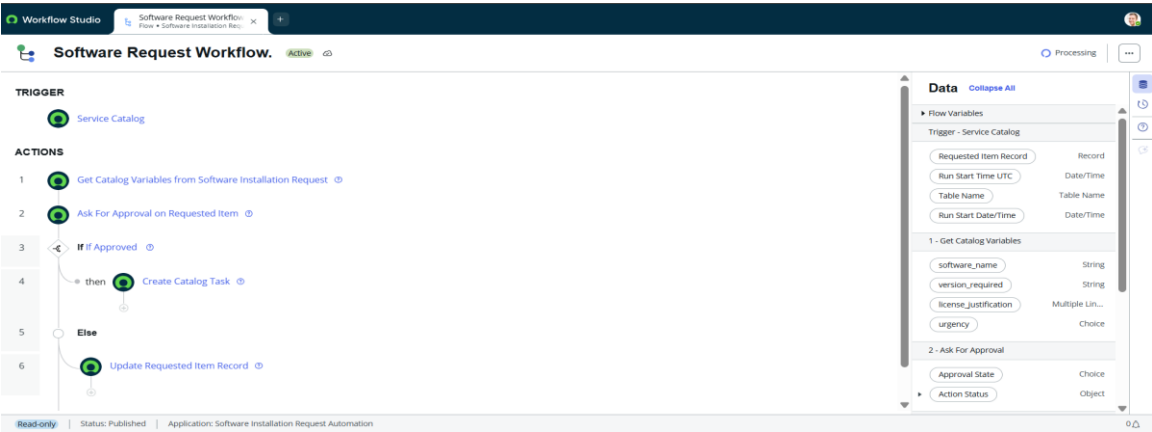
Delivery Time: 0 Days

Add to Cart

Save as Draft

Order Now

Service Portal Request Form



Flow Designer Logic

servicenow All Favorites History Workspaces Admin Catalog Task - SCTASK0010036

Search

Catalog Task SCTASK0010036

Number: SCTASK0010036

Assigned to:

Configuration item:

Active: ☒

Approval: Not Yet Requested

Priority: -- None --

State: Open

Request item: RITM0010030

Requested for: System Administrator

Variables

What software do you need?
cursor

Specify version (if required)
2.0

Why do you need this software?
programming

Select urgency level
Normal

Short description: Fulfill Software Request

Description:

Generated Catalog Task (SCTASK)

Incident - INC0010040

Number: INC0010040

* Caller: System Administrator

Category: Inquiry / Help

Subcategory: -- None --

Service:

Service offering:

Configuration item:

* Short description: License unavailable for Software Installation Request

Description: Software request could not be fulfilled due to license shortage.

Channel: -- None --

State: New

Impact: 3 - Low

Urgency: 3 - Low

Priority: 4 - Low

Assignment group: Software Support

Assigned to:

Related Search Results

Notes Related Records Resolution Information

Watch list

Work notes

Activities: 1

System Administrator

Impact: 3 - Low

Incident state: New

Opened by: System Administrator

Priority: 4 - Low

Field changes: 2026-06-27 07:41:39

Update Resolve Delete Responses

Related Links

Show SLA Timeline

Repair SLAs

Task SLAs (2) Affected CIs Impacted Services/CIs Child Incidents

Task: INC0010040

SLA definition	Type	Target	Stage	Business time left	Business elapsed time	Business elapsed percentage	Start time	Stop time
Priority 4 response (8 hours)	SLA	Response	In progress	7 Hours 1 Minute	58 Minutes	12.22%	2026-06-27 07:41:39	(empty)
Priority 4 resolution (2 day)	SLA	Resolution	In progress	1 Day 23 Hours 1 Minute	58 Minutes	2.04%	2026-06-27 07:41:39	(empty)

1 to 2 of 2

Generated License Incident

8. ADVANTAGES & DISADVANTAGES

Advantages: Vastly decreases fulfillment time, enforces standardized data intake, creates auditable approval records, and handles license exceptions proactively.

Disadvantages: Requires active ServiceNow licensing and assumes managers will interact promptly with platform notifications.

9. CONCLUSION

The project successfully delivers an enterprise-grade automated pipeline for software provisioning, resolving the core issues of manual tracking, lost approvals, and IT inefficiency.

10. FUTURE SCOPE

Future enhancements include integrating ServiceNow Software Asset Management (SAM) to perform automated, real-time license checks against the CMDB prior to generating fulfillment tasks.

11. APPENDIX

GITHUB LINK: https://github.com/dharnisha-07/Software_Installation_Request_Automation

PROJECT DEMO VIDEO LINK:

https://drive.google.com/drive/folders/14m3IDfyELQH0QGOVIuiS8C_zT8NrUxoe

Source Code (Business Rule: Create Incident on License Issue):

```
(function executeRule(current, previous /*null when async*/) {  
    var inc = new GlideRecord('incident');  
    inc.initialize();  
    inc.short_description = 'License unavailable for ' + current.cat_item.getDisplayValue();  
    inc.description = 'Software request could not be fulfilled due to license shortage.';  
    inc.caller_id = current.request.requested_for;  
    inc.assignment_group.setDisplayValue('Software Support');  
    inc.priority = 3;  
    inc.insert();  
})(current, previous);
```